## MCO-1D · PRE-FLIGHT CHECKLIST · v25

- ■ Battery ≥ 80% SOC, voltage nominal, connector secure
- ■ All 8 ODrives powered, CAN nodes 0–7 confirmed, no fault LEDs
- ■ wc_set_fuselage_width and wc_set_tpu_start set from build measurements
- ■ All 16 servos responding: mco_preflight.py --servo-check
- ■ Encoder magnets 1–8: mco_odrive_setup.py --check-magnet
- ■ TPU winglets: no cracks, slots clear, M2 bolts tight
- ■ Rope strokes inspected: no fraying, terminations secure
- ■ GS: alc_status → 3 modalities healthy, no pending tier-3
- ■ GS: envelope_status → HOVER_CAPABLE known or None (first flight)
- ■ GS: wear_status → no maintenance advisories
- ■ GS: alc_set_flight_phase pre_flight → alc_preflight_brief
- ■ RC transmitter: calibrated, all sticks centred, trims neutral
- ■ Flight area clear, wind < 8 m/s, no precipitation
- ■ FIRST FLIGHT ONLY: H-PERF-6 at top of queue (disc_registry)

# MCO-1D · EMERGENCY PROCEDURES · v25

**WING PAIR FAULT (current anomaly / unusual vibration)**

1. Check GS: resilience_status → confirm degraded_mode and active pairs
2. If ≥6 pairs active: continue with caution, monitor altitude hold
3. If <6 pairs active: reduce lever to trim detent, descend, land immediately
4. Post-flight: wear_status for winding anomaly; inspect failed pair physically

**TIER-3 SAFETY ALERT (repeated system warning)**

1. Read alert text displayed on GS
2. Take appropriate action (reduce lever, alter heading, prepare to land)
3. GS: alc_ack_alert to silence repeat (confirm you have seen it)
4. If metric degradation >15%: land at earliest opportunity

**HOVER LOST (aircraft cannot maintain altitude below 4 m/s)**

1. GS: set_hover_capable false
2. Maintain V > V_min + 0.5 m/s at all times
3. Use H-SAR-FF-1 orbit for SAR observation, H-PREC-2 for landing
4. Post-flight: alc_debrief to review envelope data

**LOW BATTERY (voltage warning on GS)**

1. Regen will reduce above 95% SOC automatically
2. At FAULT_VOLTAGE_WARN_V: begin immediate return / landing
3. At FAULT_VOLTAGE_LAND_V: land immediately, no negotiation

# MCO-1D · POST-FLIGHT LEARNING · v25

- ■ GS: alc_set_flight_phase post_flight
- ■ GS: alc_debrief (full session narrative, regen, wear, archaeology)
- ■ GS: wear_status (check hinge/stiffness/winding indices)
- ■ GS: archaeology_status (any promoted accidental improvements?)
- ■ GS: regen_status (session total and all-time recovered energy)
- ■ GS: learning_rate good/neutral/poor (pilot satisfaction signal)
- ■ If hover first confirmed today: set_hover_capable true/false
- ■ GS: persist_save_all (save genome, CMTS, ALC session, wear, regen)
- ■ Review ALC debrief: genome shifts, CMTS transfers pending?
- ■ If CMTS transfer pending: cmts_confirm or cmts_reject
- ■ Note any unusual behaviour for next pre-flight briefing
- ■ Battery: charge to storage voltage if >24h until next flight

# MCO-1D · SAR OPERATIONS · v25

**IF HOVER CAPABLE:**

Station-keep: trim detent, both pedals for lateral fine correction

Sensor pointing down: H-SP-1 (nose-down cascade)

Payload delivery: H-CR-2 kinematics, approach slowly, confirm position

Confined approach (whisper): GS whisper_on, lever 16–28% descent zone

**IF FORWARD-FLIGHT ONLY:**

Orbit target: bank 15°, V_min, nose 10° down → H-SAR-FF-1 kinematics

Area search: parallel tracks at V_min, reverse turns → H-SAR-FF-2 kinematics

Approach target: descend in orbit, nose pitch increases → H-SAR-FF-3

Payload: minimum-speed pass, drop at closest approach → H-SAR-FF-4

Landing: approach V_min, flare at 2.5m AGL → H-PREC-2

Orbit radius: $r = V^2 / (9.81 \times \tan(\text{bank_deg}))$

**THERMAL ADVISORY (H-SOR-6):**

System will advise bearing, bank angle, and turn direction. YOU decide whether to turn. System NEVER turns autonomously. GS: thermal_advisory_status for current reading.

# MCO-1D · ALC QUICK REFERENCE · v25

**STERILE COCKPIT — ALC is SILENT during: launch, takeoff, climb, manoeuvre, approach, landing**

**ALC SPEAKS ONLY during: cruise (brief obs), pre-flight brief, post-flight debrief, tier-3 alerts**

| Command | When | Purpose |
|---|---|---|
| `alc_preflight_brief` | Before flight | What was learned, what is queued |
| `alc_debrief` | After landing | Full session summary |
| `alc_status` | Any time | Phase, alerts, modalities |
| `alc_set_flight_phase X` | Phase change | pre_flight / cruise / approach / post_flight |
| `alc_ack_alert` | Tier-3 fires | Acknowledge safety alert |
| `learning_rate good` | Any time | Reinforce current config in genome |
| `learning_rate poor` | Any time | Reduce current config weight |
| `wear_status` | Post-flight | Per-wing mechanical health |
| `regen_status` | Post-flight | Session and all-time energy recovered |
| `envelope_status` | Any time | HOVER_CAPABLE, V_min, orbit radius |
| `set_hover_capable true` | Post first flight | Confirm hover capability |
| `set_hover_capable false` | Post first flight | Confirm forward-flight only |
| `persist_save_all` | End of session | Save all learning to disk |